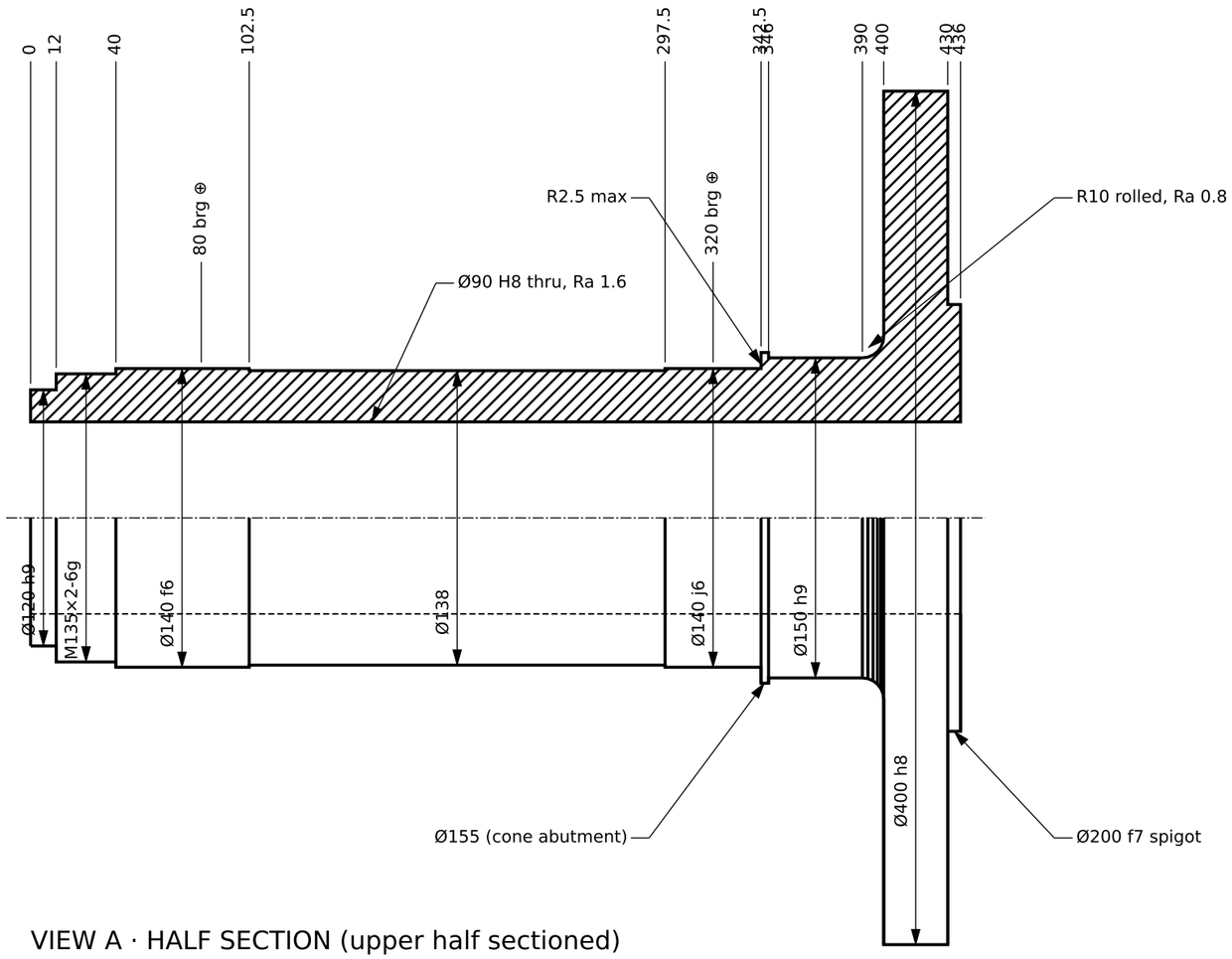
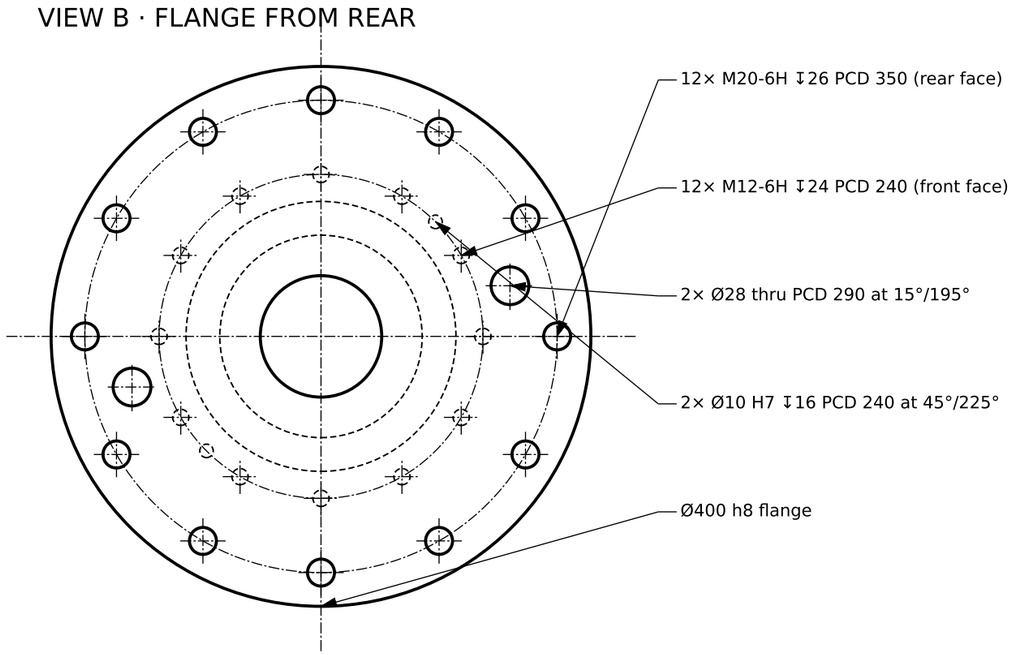


VIEW A · HALF SECTION (upper half sectioned)



VIEW B · FLANGE FROM REAR

NOTES

1. Blank: open-die forging 42CrMo4, forging ratio ≥ 3, quenched and tempered to Rp0.2 ≥ 650 MPa, Rm 900 to 1100 MPa; KV ≥ 27 J at −40 °C (3 specimens, longitudinal, taken from the flange zone). Certificate EN 10204 3.1.
2. Ultrasonic test of the blank to EN 10228-3 quality class 3; MT of the finished part (fillets, shoulders, thread) to EN ISO 9934, no linear indications.
3. Bearing seats: Ø140 f6 front (cone must slide for preload), Ø140 j6 rear; Ra 0.8; cylindricity 0.006; run-out of both seats and of Ø400 and Ø200 to common axis A-B ≤ 0.01.
4. Flange rear face and front face perpendicular to A-B within 0.02; flatness 0.02. Ø400 seal surface Ra 0.2 to 0.4, plunge ground (no lead), hard chrome 30 µm or wear sleeve.
5. Thread M135×2-6g with undercut DIN 76-D; precision locknut KMT 27 (or equivalent with locking screws).
6. Bore Ø90 H8 for the cooling insert NP1-110 (installed with liquid nitrogen shrink, interference 0.02 to 0.06).
7. Two radial holes Ø6 from the bore to under each bearing cone seat at 90° (bottom dead centre at 0°) for PT100 bearing temperature sensors; deburr, edges R0.5.
8. All fillets and the shoulders rolled or shot peened; no stamping on the part: mark serial number on the flange OD rear chamfer by etching.
9. Protect: phosphate + oil in storage; seats and seal surfaces taped until assembly.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

SPINDLE (STATIONARY MAIN SHAFT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-100	A · 2026-11-02	1:2.86 (A) / 1:4.5 (B)
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
42CrMo4+QT (EN 10083-3), forged	56.7 kg	Claude / not checked